Standard Operating Procedure for Resolving Routing Protocol Flap on KR-PE-15 at KR Puram Access

Node: KR-PE-15 | **Vendor:** Cisco | **Severity:** Minor | **Alarm Code:** ALARM_7942

Description

A Routing Protocol Flap on KR-PE-15 at KR Puram Access occurs due to excessive BGP route churn, OSPF adjacency loss, or ISIS LSP flooding, likely resulting from a combination of factors including misconfigured routing policies, hardware malfunctions, or software bugs. The PE router's control plane is heavily reliant on its ability to maintain stable routing protocol adjacencies, which are crucial for efficient data forwarding. A minor alarm (ALARM_7942) has been raised, but the impact on the network is minimal. To prevent further disruptions, it is essential to identify the root cause and swiftly implement corrective actions.

Pre-requisite Checks

- Verify that the KR-PE-15 router is accessible remotely.
- Check the status of the redundant links at KR Puram Access to ensure they are operational.
- Confirm that the power supply to the router is stable and not affected by any issues.

Potential Root Causes

Cause	Probability
Misconfigured BGP routing policies	High
Failed or malfunctioning hardware components	Medium
Software bugs or version incompatibilities	Low

Diagnostic Commands & Expected Outputs

```
Command: show ip bgp summary, Expected Output: BGP table version is 10, local router ID is 192.168.1.1, Description: This command displays a summary of the BGP table, including the number of prefixes and the router ID. It helps in identifying if there are any issues with the BGP table.

Command: show ip ospf neighbors, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: This command displays information about OSPF neighbors, including their state and dead time. It helps in identifying if there are any issues with OSPF adjacencies.

Command: show isis database, Expected Output: ISIS LSP Database, Description: This command displays the ISIS LSP database, which includes information about the LSPs and their sequence numbers. It helps in identifying if there are any issues with ISIS LSP flooding.
```

Resolution Steps

Step 1: Step: Reset the BGP session, Command: clear ip bgp *, Description: This command clears the BGP session and resets the BGP table. It helps in resolving issues related to misconfigured BGP routing policies.

Step 2: Step: Replace failed hardware components, Command: no shutdown, Description: This command enables the interface and brings it up. It helps in resolving issues related to failed or malfunctioning hardware components.

Step 3: Step: Upgrade or downgrade to a compatible software version, Command: archive download-sw /force-reload, Description: This command downloads the new software image and reloads the router. It helps in resolving issues related to software bugs or version incompatibilities.

Post-Check Verification

```
Command: show ip bgp summary, Expected Output: BGP table version is 10, local router ID is 192.168.1.1, Description: This command displays a summary of the BGP table, including the number of prefixes and the router ID. It helps in verifying if the corrective actions were successful.

Command: show ip ospf neighbors, Expected Output: Neighbor ID Pri State Dead Time Address Interface, Description: This command displays information about OSPF neighbors, including their state and dead time. It helps in verifying if the corrective actions were successful.

Command: show isis database, Expected Output: ISIS LSP Database, Description: This command displays the ISIS LSP database, which includes information about the LSPs and their sequence numbers. It helps in verifying if the corrective actions were successful.
```

Roll-back Procedures

- Step: Restore the previous BGP configuration, Command: copy run start, Description: This command saves the current configuration and reloads the router. It helps in restoring the previous BGP configuration in case the corrective actions fail.

- Step: Replace the hardware components, Command: no shutdown, Description: This command enables the interface and brings it up. It helps in replacing the hardware components in case the corrective actions fail.

- Step: Downgrade to the previous software version, Command: archive download-sw /force-reload, Description: This command downloads the previous software image and reloads the router. It helps in downgrading to the previous software version in case the corrective actions fail.